# Shardul Kulkarni

Founding AI Engineer | Agentic AI / GenAI | Enterprise AI Delivery

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## SUMMARY

AI engineer and delivery leader with 3+ years taking GenAI and Agentic AI systems from discovery to production for enterprise clients. As Founding AI Engineer and later CTO of MAindTec GmbH (to September 2026) I owned the full lifecycle: scoping business problems with customers, architecting scalable solutions, shipping production systems on Azure and AWS, and driving measurable outcomes. Delivered 3 enterprise AI programmes across manufacturing and automotive and led a team of 9 engineers. Available immediately.

## SKILLS

**GenAI / Agentic AI:** LangGraph, LangChain, Claude Agent SDK, MCP Servers, Multi-Agent Orchestration, Tool Calling, Structured-Output Validation, RAG (hybrid search, reranking), Prompt and Context Engineering, LLM Fine-tuning, Evaluation (LangSmith), LLMs (OpenAI, Anthropic, HuggingFace)

**AI / ML Engineering:** PyTorch, LightGBM, Computer Vision (YOLO, SegmentAnything, OpenCV), Object Detection and Tracking, Model Quality Metrics, Production Monitoring, LLMOps / MLOps

**Engineering / Backend:** Python (5+ years), FastAPI, Pydantic, pytest, TypeScript, Node.js, Next.js 15, REST APIs, PostgreSQL, pgvector, Milvus, Redis

**Cloud / DevOps:** Azure (Container Apps, OpenAI Service, Blob, Key Vault, Bicep IaC), AWS (Bedrock, customer-cloud deployment), Docker, CI/CD (GitHub Actions, Azure DevOps), Linux Server Administration

**Consulting / Delivery:** Enterprise Client Delivery, Stakeholder Management (C-level), Use Case Workshops, Technical Demos and Pre-Sales, Solution Architecture, AI Adoption Strategy, Team Leadership, Industry Fair Presentations

## EXPERIENCE

**MAindTec GmbH (B2B AI SaaS Startup)**, Ingolstadt, Germany

*Founding AI Engineer (formerly Chief Technology Officer)* · January 2026 - September 2026

- Led end-to-end delivery of GenAI and Agentic AI programmes for enterprise clients in manufacturing and automotive, from discovery and solution design through production rollout and adoption.
- Built and shipped MAiQ, a production multi-agent LLM platform: LangGraph agents, tool calling, MCP servers, structured-output validation, hybrid RAG with citations (pgvector, BM25 + dense retrieval, Cohere reranking) and LangSmith evaluation and observability; multi-tenant with RBAC and SAML SSO on Azure (FastAPI, Next.js 15).
- Scaled SpendSense, an ML procurement price-benchmarking and cost-optimisation product (LightGBM, Python), from proof of concept to production flagship; it won a major enterprise project.
- Attended investor pitches and contributed to a strategic investment from the Tragus Group; led MAindTec's accreditation into Microsoft for Startups (Founders Hub).
- Acted as the bridge between engineering, C-level stakeholders and business users: discovery workshops, technical demos, architecture decisions and AI adoption strategy.
- Delivered 3 enterprise AI programmes end-to-end from discovery through deployment; led a team of 9 engineers across all functions.

*Lead Solution Engineer, AI and DevOps* · June 2025 - December 2025

- Delivered end-to-end AI solutions for enterprise clients; led technical demos, scoping sessions and architecture decisions.
- Built AI for engineering drawings: automated analysis of 2D and 3D CAD and JT files for manufacturing and automotive clients.
- Stood up self-hosted vector search at scale (Milvus) for retrieval across large enterprise datasets.

*AI and Software Engineer* · April 2025 - May 2025

- Delivered the company's first client project, an AI solution for the City of Ingolstadt, to production.
- Built RAG-powered AI applications with LangChain and FastAPI; first production deployment on Azure with CI/CD.

**Otto-von-Guericke University**, Magdeburg, Germany

*Research Assistant, Generative AI* · December 2024 - March 2025

- Built RAG pipelines with agent-based retrieval to interpret medical workflow specifications and documents.

**mediMESH, Clinical Insights**, Magdeburg, Germany

*AI and Software Engineer (Working Student)* · March 2023 - March 2025

- Built production computer-vision systems in PyTorch over terabyte-scale clinical video: custom YOLO detection, SegmentAnything, multi-camera multi-object tracking.
- Engineered a GPU-accelerated anonymisation pipeline with human review in CVAT, cutting processing from 3 days to under 7 minutes (99.84%) for GDPR-compliant analytics.

## PROJECTS

**MAiQ — AI Engineering Platform** · *Production B2B SaaS, Enterprise Clients*

- Enterprise AI platform: LangGraph agents, hybrid RAG with citations, MCP servers, LangSmith observability. Multi-tenant, RBAC, SAML SSO. Next.js 15, FastAPI, pgvector, Azure. In production for manufacturing and automotive clients.

**SpendSense — AI Price Benchmarking** · *Production, Predictive ML, MAindTec GmbH*

- LightGBM procurement price-benchmarking and cost-optimisation product; grew from proof of concept to production flagship and won a major enterprise project.

**honestapply** · *Open source (AGPL), github.com/shardulkulkarni14/honestapply*

- Local-first autonomous job-application agent: discovers roles, scores fit, tailors CV and cover letter, fills forms via a browser agent; enforced no-fabrication checks, 187 tests, multi-provider (Anthropic, OpenAI-compatible, Gemini).

## EDUCATION

**M.Sc. Data and Knowledge Engineering · 2022 - 2024**

Otto von Guericke University, Magdeburg, Germany · *GPA: 1.7; thesis grade 1.3*

**B.E. Computer Engineering · 2017 - 2021**

Savitribai Phule Pune University, Pune, India · *GPA: 1.4*

## AWARDS

- Smart India Hackathon 2020, Top 5 Finalist (Government of India, 10,000+ teams)
- Two published Indian patents (2022): Smartzzy (App. 202121011507) and Smart Parkade (App. 202121011511)

## LANGUAGES

English (C1) · German (A2, improving) · Hindi (Native) · Marathi (Native)